



PharmaTrack: A Web-Based Inventory Management System for Tracking Medicine Expiration and Stock in Pharmacies

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Jamaica Ruri C. Augusto
Bachelor of Science in Computer Science

Mr. Mark Anthony Cabilo
Special Problem Adviser

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Chapter 1. Introduction

1.1 Background of the Study

Pharmacies play a crucial role in healthcare by ensuring that medicines are available, safe, and effective for patient use. However, many small and medium-sized pharmacies still rely on manual or semi-digital methods, such as logbooks or spreadsheets, to manage medicine inventory and expiration dates. These methods are prone to human error, delayed updates, and overlooked expired medicines, which may lead to financial losses and potential risks to patient safety.

While Point of Sale (POS) systems are commonly used in pharmacies to handle sales transactions, not all pharmacies utilize advanced inventory features, and many are unwilling or unable to integrate external systems due to operational risks, privacy concerns, or testing limitations. During the conduct of this study, no pharmacy was available to serve as a live testing environment for system integration. As a result, this research focuses on the development and evaluation of a standalone, web-based pharmacy inventory and expiration monitoring system tested in a simulated environment.

A recurring problem observed in small and medium-sized pharmacy settings is that pharmacists and pharmacy staff experience significant difficulty in monitoring medicine stocks and expiration dates efficiently. When relying on manual methods, staff are required to individually look up each medicine record to determine whether a product is running low in quantity or approaching its expiration date. This process is time-consuming, susceptible to oversight, and places an unnecessary burden on pharmacy staff who must perform these checks alongside their other daily responsibilities. The absence of an automated monitoring system means that expired or critically low-stock medicines may go unnoticed until they become an operational or patient safety issue.

This study proposes PharmaTrack, a web-based inventory management system designed to directly address this problem by automating the monitoring of medicine stock levels and expiration dates without requiring direct integration with an existing POS system. The system allows manual and bulk entry of medicine records into a structured database and automatically generates reports for medicines that are low in stock or nearing expiration, eliminating the need for staff to perform manual lookups. While existing studies on pharmacy inventory systems have explored digital record keeping and automated alerts, most focus on enterprise-level or POS-integrated solutions that are impractical for smaller pharmacies, and few have addressed the specific challenge of transitioning existing manual or spreadsheet-based records into a digital system with minimal disruption. PharmaTrack addresses this gap by introducing a bulk data import feature that allows administrators to upload medicine records directly from a CSV file, enabling pharmacies to migrate their existing spreadsheet data into the system efficiently and without requiring technical expertise. By focusing on essential inventory functions and providing a practical migration pathway from manual records, PharmaTrack offers a low-risk solution

suitable for small and medium pharmacies seeking to improve the efficiency of their inventory monitoring without modifying their current sales systems.

1.2 Research Objectives

The main objective of this study is to develop and evaluate PharmaTrack, a standalone web-based pharmacy inventory management system adapted from an open-source application, assessing whether the system improves the efficiency of pharmacists and pharmacy staff in monitoring medicine stock levels and expiration dates compared to manual tracking methods.

Specifically, the study aims to:

- Evaluate the efficiency of the system by measuring the effectiveness and task completion time of pharmacists and pharmacy staff when using PharmaTrack to perform inventory monitoring tasks, in comparison to a simulated manual tracking condition.
- Develop a system that supports two user roles, administrator and pharmacist, with clearly defined access permissions.
- Enable administrators to add medicines individually or import bulk medicine records using a CSV file-based data migration approach.
- Provide real-time visibility of medicine inventory, including quantities and expiration dates, through automated monitoring and report generation.
- Generate inventory reports related to near-expiry medicines and low-stock quantities to replace the need for manual record lookups.
- Allow pharmacists to efficiently view and search inventory records and access inventory reports without modification privileges.

1.3 Significance of the Study

The study will benefit the following stakeholders:

Pharmacists and Pharmacy Staff – By reducing manual tracking errors and providing timely reports, the system helps them manage stocks more efficiently and avoid dispensing expired medicines.

Pharmacy Owners – Accurate inventory data prevents overstocking or understocking, reducing losses and improving profitability.

Customers/Patients – Ensures that only safe and effective medicines are available, improving patient trust and safety.

Future Researchers – The study can serve as a reference for those exploring pharmacy management systems, digital inventory solutions, or healthcare IT systems.

1.4 Scope and Limitation

Scope:

The study focuses on the development of PharmaTrack, a web-based pharmacy inventory management system with role-based access control. The system supports two user roles: administrator and pharmacist. Administrators have full access to inventory management features, including adding new medicines, importing bulk medicine records, editing or deleting records, and viewing inventory reports. Pharmacists are limited to viewing inventory records through a search interface and accessing reports related to expiration and quantity.

The system includes a bulk data migration feature that allows administrators to import medicine records using a structured file format. The system is evaluated in a simulated environment using sample datasets.

Limitations:

- The system is not integrated with any existing pharmacy POS system or live operational database.
- The evaluation is limited to simulated data and user-based testing rather than real-world pharmacy deployment.
- The bulk import feature supports only a predefined file format and does not include automatic data validation against external pharmacy systems.
- The system does not cover other pharmacy operations such as billing, insurance claims, prescription handling, or customer management.
- Advanced tracking technologies such as RFID, barcode scanners, or mobile device support are not included.
- The system is designed for desktop or laptop use only.

1.5 Definition of Terms

Administrator – A user with full access rights to manage inventory data, import records, and generate reports.

Inventory Management – The process of overseeing the ordering, storage, and use of medicines in a pharmacy.

Pharmacist – A user with restricted access, limited to viewing inventory records and reports.

PharmaTrack – The proposed system that monitors medicine stocks and expiration dates in pharmacies.

POS (Point of Sale) – A system used to process sales transactions at the counter but not necessarily for inventory or expiration monitoring.

Role-Based Access Control – A security mechanism that restricts system access and functionalities based on user roles such as administrator and pharmacist.

Web-Based Application – Software that runs on a web browser, accessible via internet-connected devices.

Chapter 2. Review of Related Literature

2.1 Pharmacy Inventory Management Systems

Effective inventory management is fundamental to pharmacy practice because medicines represent one of the largest and most dynamic assets of a pharmacy. Studies and reviews consistently show that inventory management goals include balancing availability with cost, minimizing both stock-outs and wastage due to expiry, while also maintaining traceability of batches and lot numbers. Literature reviews in pharmacy practice emphasize that poor inventory control increases operational costs, reduces service quality, and can directly threaten patient safety when expired or improper stock is dispensed [1].

The literature also demonstrates that inventory challenges vary across settings. Hospital inpatient pharmacies face unique requirements, such as high turnover, emergency drug availability, and strict ward dispensing flows. In contrast, community pharmacies typically manage a wide range of stock keeping units (SKUs) in lower volumes with more variable demand patterns. The chapter on pharmaceutical inventory and analysis in Springer notes that hospital systems often employ structured classification and prioritization of medicines to ensure timely availability for critical care, but these approaches are not always feasible or cost-effective for smaller community pharmacies [2].

Local studies on information system design for pharmacy businesses highlight that a systems approach, such as combining ABC/XYZ or ABC-XYZ analysis with digital record keeping, can significantly improve reorder decisions and reduce expiry losses. Case studies report measurable improvements in stock turnover and fewer expired items after the adoption of computerized inventory tools [3], [4]. These findings indicate that even limited digitalization, such as basic data entry with automated checks, can outperform purely manual recording.

Despite these benefits, the literature also notes barriers to adoption. High implementation costs, staff training requirements, workflow disruptions, and system complexity remain key obstacles. Many small pharmacies continue to rely on manual records or simple spreadsheets because commercial pharmacy systems are often perceived as expensive or over-featured. This gap suggests an opportunity for simpler web-based systems that focus on the core inventory challenges of stock quantity and expiry monitoring, without requiring full enterprise resource planning (ERP) or point of sale (POS) functionality [1]–[4]. PharmaTrack is positioned to address this gap by offering a focused and accessible web solution for stock and expiry monitoring.

2.2 Point of Sale (POS) Systems in Pharmacies

Point of Sale (POS) systems are widely used in retail and pharmacy settings for transaction recording, receipt generation, and basic stock decrementing on sales. The literature on POS adoption in pharmacies indicates that POS improves transactional efficiency and provides sales reports, but conventional POS platforms often lack batch-level tracking and automated expiry monitoring [5]. As a result, pharmacists frequently maintain separate manual logs to track expiries and restocking needs.

Empirical studies and practitioner reports, including research conducted in Southeast Asia, show that the typical POS workflow subtracts a unit when a sale is made but rarely ties that sale to a specific batch or expiration date. Consequently, while POS systems can provide aggregate consumption data, they do not inherently prevent the sale of a near-expiry lot unless

combined with additional inventory modules or manual procedures [6]. This separation creates a blind spot in expiry control for many small and independent pharmacies.

Operations research and systems design literature emphasizes integrating POS with inventory control to produce traceable, batch-aware records. Several studies propose architectural or process modifications, such as augmenting POS workflows with batch selection prompts or integrating batch import tools so that incoming deliveries are recorded with batch and expiry metadata at the time of receipt [7]. However, such integrations often require vendor cooperation or additional middleware, which increases cost and complexity.

Since many small pharmacies cannot afford fully upgraded POS or integrated solutions, lightweight web applications that provide importable inventory tables, expiry checks, and alerts can complement existing POS setups. A focused tool that is interoperable through CSV imports and exports is often more feasible and practical than a complete POS replacement, which is precisely the niche PharmaTrack seeks to serve.

2.3 Advantages of Expiration Tracking Systems

Medical and management literature highlights that proactive expiration tracking has multiple benefits: it reduces clinical risk by preventing expired medicines from reaching patients, decreases economic loss by enabling stock rotation or discounted use before expiry, and improves regulatory compliance by maintaining audit trails for disposal. Quantitative and qualitative studies report substantial reductions in expired stock and associated financial losses when automated expiry monitoring systems are adopted [8].

Technical literature also describes different mechanisms for expiry management, including scheduled processes that flag near-expiry lots, dashboard visualizations that display “expiring soon” reports, and automated notification channels such as email or push alerts. Comparative studies suggest that systems combining visibility through dashboards with automated reminders are most effective in changing daily behavior and reducing expiry incidents, as they reduce reliance on manual record scanning [9].

Beyond preventing losses, expiry tracking supports procurement decisions. Systems that aggregate expiry dates across batches and SKUs allow purchasing decisions to prioritize high-turnover medicines while delaying replenishment of those with long remaining shelf life. This practice reduces both wastage and tied-up capital. Supply-chain and computing literature outlines how combining expiry metadata with forecasting models enhances inventory turnover and lowers carrying costs [10].

Nevertheless, the literature also warns of implementation challenges. Poor data quality during entry, insufficient staff training, and lack of integration with procurement workflows can reduce the benefits. For this reason, simple, well-designed interfaces for reliable manual entry and bulk import functionality are recommended. These insights inform PharmaTrack’s design choices, which prioritize minimizing data-entry friction, validating expiry dates at input, and providing bulk import tools so pharmacies can transition from spreadsheets with minimal disruption.

2.4 Security and Data Privacy in Pharmacy Systems

As pharmacy systems become digital, the importance of information security and data privacy rises accordingly. Pharmacy information systems may capture supplier details, pricing, transaction logs, and if extended in the future, patient or prescription identifiers. Academic reviews

and policy reports emphasize that weak security in health and pharmacy systems risks data breaches, regulatory non-compliance, and erosion of trust among stakeholders [11].

Several studies and policy analyses point to core security controls that are applicable even to inventory-only systems: strong authentication (role-based access control), encrypted transport and storage of sensitive fields, audit logs for critical updates (for example, expiration date edits), and secure backups. The literature further emphasizes that security is not an add-on but must be designed into the system from the start to prevent misconfigurations that could expose data [12].

Legal and regulatory frameworks (for example HIPAA in the United States, or national data privacy laws like the Philippines' Data Privacy Act) shape required controls if systems process personal health information. Even if a system intentionally avoids collecting patient data (as PharmaTrack does in its core scope), the literature recommends adopting security best practices because supplier and transactional data still have business value and may be subject to regulation or contractual confidentiality [13], [14]. Beyond these frameworks, broader research on health information management highlights that both nursing and pharmacy professionals face shared challenges in protecting sensitive data, particularly in the context of digital transformation and cross-disciplinary workflows [15].

Finally, usability research in health IT underlines a trade-off: security measures that are too cumbersome (e.g., frequent forced password changes, complex multi-factor flows) can push staff to adopt insecure workarounds (shared credentials, paper backups). The recommended approach is to balance security with usability: role-based permissions, session timeouts, and optional multi-factor authentication for admin roles provide strong protection without crippling daily tasks.

2.5 Current Solutions and Gaps

The commercial market for pharmacy systems includes both enterprise-grade solutions (such as hospital ERP modules) and smaller POS or inventory applications. Enterprise systems offer end-to-end functionality, including procurement, batch traceability, physician order integration, and disposal workflows. However, they are costly, complex, and generally designed for hospitals and large pharmacy chains [17]. By contrast, many POS and lightweight inventory tools primarily focus on sales and stock decrementing but lack robust expiry monitoring and batch handling, features that are crucial for minimizing expiry-related risks [16], [18].

Studies on pharmacy automation consistently reveal that functionality critical to preventing expiry, such as batch entry at receipt, batch-aware dispensing, and expiry-based reporting and alerts, is typically available in higher-end solutions but either missing or poorly implemented in mass-market tools. Furthermore, vendor lock-in and high integration costs make it difficult for smaller pharmacies to adopt these advanced systems even when beneficial.

This situation presents a clear opportunity for innovation. A focused, web-based application that is affordable, easy to use, and tailored to community pharmacy workflows could address this gap. PharmaTrack is designed to provide: (1) search function for easy findings of data; (2) simple data input and bulk import through CSV/Excel, enabling a smooth transition from manual records; and (3) automated expiration checks with configurable alerts. By keeping the scope tight and the interface intuitive, PharmaTrack aims to lower adoption barriers and deliver high-impact features needed by smaller pharmacies.

The literature consistently shows that inventory management and expiry control are essential to pharmacy operations for reasons of safety, compliance, and financial efficiency. While enterprise and hospital systems offer robust capabilities, many small and community pharmacies still rely on manual methods or basic POS tools that do not adequately support batch-level tracking or automated expiry alerts. Studies recommend practical solutions that combine simple digital entry, bulk import, automated checks, and usable dashboards. Security and data privacy considerations are also crucial but manageable if best practices are followed without overburdening staff. Overall, there exists a clear research and practical niche for a focused, web-based system such as PharmaTrack, which prioritizes manual/import data entry, search functionality, and effective expiration reports suited to the realities of small pharmacy operations.

Chapter 3. Methodology

3.1 Research Design

The open-source pharmacy management system developed by Varshini Elangovan (available at her GitHub repository, github.com/Varshini-E/Pharmacy-Management-System) was selected as the base system for several reasons. First, its technology stack, which consists of PHP, MySQL, HTML5, CSS3, and JavaScript, is aligned with the requirements of this study, enabling straightforward modification without introducing new dependencies. Second, the original system's existing modules for medicine inventory and role-based access control provide a functional foundation upon which the study's enhancements, specifically bulk CSV data migration and expiration monitoring, could be meaningfully built. Third, as a publicly available open-source project with clear and verifiable authorship, proper attribution is academically defensible and traceable. The adaptation approach follows a developmental research design, wherein the research contribution lies in the targeted enhancements made to address the identified gaps in the original system. Notably, the bulk data import feature using CSV files was not present in the original system and was developed specifically for this study, along with the removal of modules unrelated to inventory and expiration tracking, the expansion of medicine datasets, and the redefinition of user roles to reflect real pharmacy workflows.

Several modules from the original system that were outside the scope of this study were removed, specifically the Suppliers module, Stock Purchase module, Customer Management module, and Sales-related modules including the sales invoice details, sold product details, and the add new sale function. These removals were deliberate, as the study focuses exclusively on inventory monitoring and expiration tracking rather than transactional or customer-facing operations. The retained modules are the Dashboard, Inventory, and Reports, to which the Bulk Import module was added as the primary research contribution of this study.

3.2 System Overview

PharmaTrack is a standalone, web-based pharmacy inventory management system adapted from an existing open-source pharmacy management application developed using PHP and MySQL. The system is customized to focus on inventory tracking, expiration monitoring, and controlled data access through user roles.

The system supports two user roles: administrator and pharmacist. Administrators are responsible for managing employees and medicine records, including adding new medicines manually or importing bulk medicine records through file upload. Pharmacists are provided read-only access, allowing them to search and view inventory data and generate inventory reports without modifying records.

Bulk data migration is supported using a structured file format, enabling efficient population of the database with large sets of medicine records. The system processes the

uploaded file, validates required fields, and inserts records into the database. Inventory data is continuously evaluated to identify medicines that are low in stock or nearing expiration, with results presented through reports.

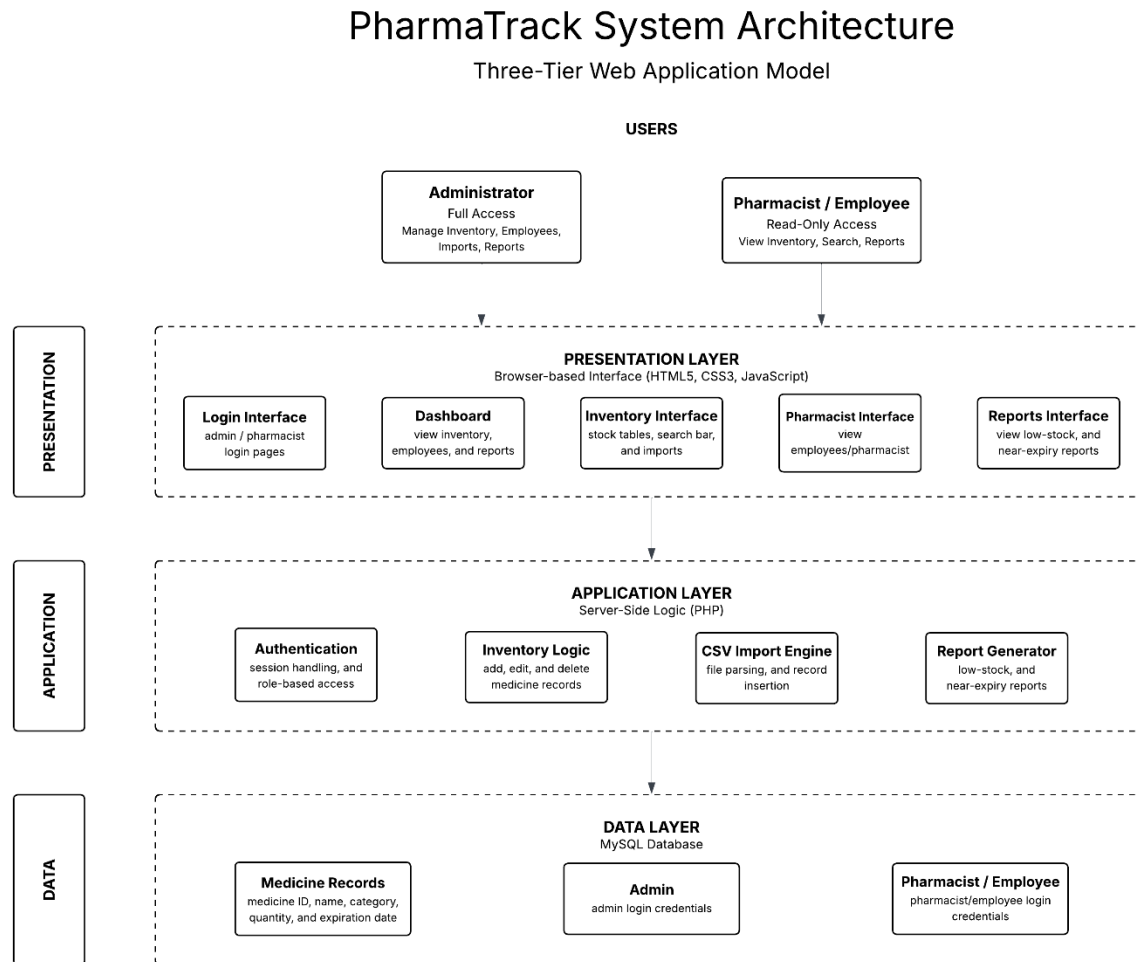


Figure 1: System Architecture

3.3 Tools and Technologies

The system is implemented using a traditional web development stack commonly used in small-to-medium scale information systems. The technologies used are based on the original open-source implementation and were retained to ensure compatibility and ease of modification.

PHP serves as the primary server-side scripting language of the system. It handles the core system logic, including inventory processing, expiration checks, report generation, and all interactions with the database. PHP scripts are responsible for processing user requests and dynamically generating web content in response.

MySQL serves as the Database Management System for storing all system data, including medicine inventory records, expiration dates, and alert logs. For this study, additional medicine datasets were added to the database to simulate real pharmacy inventory conditions. HTML5 is used to structure the web pages of the system, defining the layout of forms, tables, dashboards, and system notifications displayed to users. CSS3 complements this by providing styling for tables, alerts, buttons, and page layouts, ensuring the usability and readability of inventory information across the interface. JavaScript is used to improve interactivity and user experience by supporting client-side validation, dynamic updates to tables, and visual alert indicators for low-stock and near-expiration medicines.

For bulk data migration, the system supports the Comma-Separated Values (CSV) file format. CSV files are lightweight, widely supported, and compatible with spreadsheet applications such as Microsoft Excel and Google Sheets. This format allows administrators to prepare and migrate large datasets efficiently while maintaining a simple structure suited for server-side parsing using PHP.

3.4 System Architecture

The architecture of PharmaTrack follows a three-tier web application model as illustrated in Figure 1. The system is structured into three distinct layers: the Presentation Layer, the Application Layer, and the Data Layer. Each layer has a clearly defined responsibility, and communication flows sequentially from the user interface down to the database, ensuring separation of concerns and maintainability of the system.

3.4.1 Presentation Layer

The Presentation Layer serves as the interface between the users and the system. It is developed using HTML5, CSS3, and JavaScript, and is accessed entirely through a web browser, requiring no additional software installation on the user's device. This layer renders all visual components that users interact with, including forms, tables, dashboards, and alert indicators.

The layer is composed of five main interface components. The Login Interface provides separate entry points for the Administrator and the Pharmacist, enforcing role-based access from the moment a user enters the system. The Dashboard provides a centralized view of inventory, employee, and report icon buttons, serving as the primary navigation hub after login. The Inventory Interface displays stock tables, a search bar for quick record retrieval, and the bulk import form that allows administrators to upload CSV files. The Pharmacist Interface allows the administrator to view and manage pharmacist and employee records. Finally, the Reports Interface presents generated reports on low-stock medicines and near-expiry items in a readable tabular format.

JavaScript handles client-side validation and dynamic interface updates, such as highlighting medicines that are flagged as near-expiry or below the stock threshold, reducing the need for full page reloads when displaying alert statuses.

3.4.2 Application Layer

The Application Layer contains the core business logic of PharmaTrack and is implemented entirely in PHP. This layer processes all requests received from the Presentation Layer, applies the appropriate logic, and communicates with the Data Layer to retrieve or update records. It acts as the intermediary between what the user sees and what the database stores.

Access control is enforced at this layer through the Authentication module, which manages user sessions and verifies role permissions on every request. This ensures that pharmacists cannot access administrator-exclusive functions such as record editing, bulk import, or employee management, even if they attempt to navigate directly to those pages through the browser.

The Inventory Logic module handles all create, read, update, and delete operations for medicine records. It processes form submissions from the Presentation Layer and translates them into database queries. The CSV Import Engine is a feature developed specifically for this study and was not present in the original open-source system. It receives uploaded CSV files, parses each row, validates the required fields, and inserts the records into the database in bulk, enabling efficient data migration from spreadsheets. The Report Generator module queries the database for medicines whose expiration dates fall within a predefined threshold or whose stock quantities fall below a set minimum, and prepares the results for display in the Reports interface.

3.4.3 Data Layer

The Data Layer consists of a MySQL relational database that serves as the persistent storage for all system data. It is the only layer that directly reads from and writes to the database, as all data requests are routed through the Application Layer.

The database contains three primary data stores relevant to the adapted system. The Medicine Records table is the central table of the system, storing each medicine's ID, name, category, quantity on hand, and expiration date. This table is the primary source for both inventory display and report generation. The Admin table stores the login credentials exclusive to the administrator account, and access to this table is restricted to the authentication process only. The Pharmacist/Employee table stores the login credentials for all pharmacist and employee accounts, and is similarly accessed only during the authentication process to verify user identity and assign the appropriate role-based session.

Together, the three layers operate as a cohesive system where each tier depends on the one below it but remains independent enough to be modified without breaking the others. This separation ensures that changes to the interface do not require changes to the database structure, and that business logic remains centralized in the Application Layer rather than being scattered across the interface or the database.

3.5 System Features

The adapted PharmaTrack system is built around four core modules: the Dashboard, Inventory, Reports, and Bulk Import. Access to these modules is governed by a role-based access control mechanism that defines two distinct user roles. The Administrator is granted full access to

all system functions, including inventory management, employee management, bulk data import, record editing and deletion, and report generation. The Pharmacist is provided with read-only access, limited to viewing and searching inventory records and accessing generated reports without the ability to modify any data.

The Inventory Management feature allows administrators to add, edit, and delete individual medicine records, each containing details such as medicine name, category, quantity on hand, and expiration date. Complementing this is the Bulk Medicine Record Import feature, which was developed specifically for this study and was not present in the original open-source system. It allows administrators to upload a structured CSV file containing multiple medicine records, which the system then parses, validates, and inserts into the database, enabling efficient migration from existing spreadsheets.

The system also includes automated monitoring capabilities. The Expiration Monitoring feature continuously evaluates medicine records and identifies those whose expiration dates fall within a predefined threshold, while the Quantity Monitoring feature flags medicines whose stock levels drop below a set minimum. Both conditions are surfaced through the Reports feature, which generates structured reports on near-expiry and low-stock medicines accessible to both user roles. A Search Functionality is likewise available within the Inventory module, enabling quick retrieval of specific medicine records through a search bar.

3.6 Development Procedure

The development of PharmaTrack followed a sequential procedure designed to ensure that the adapted system aligned with the objectives of the study at every stage. The process began with a requirement analysis, during which the inventory, role-based access, and reporting requirements of the system were identified and defined. This was followed by a thorough review of the original open-source pharmacy management system developed by Varshini Elangovan, which served as the foundation for the adaptation. Once the source code was sufficiently understood, system customization was carried out by removing modules unrelated to the study scope, specifically the Suppliers, Stock Purchase, Customer Management, and Sales modules, and by implementing role-based access control to differentiate administrator and pharmacist functions. Following this, the Bulk Import feature was developed, introducing a CSV file upload and parsing mechanism that allowed large sets of medicine records to be migrated into the database efficiently. The database was then enhanced by populating it with both imported and manually entered medicine datasets to simulate realistic pharmacy inventory conditions. With the backend logic in place, interface adjustments were made to reflect the access levels of each user role, ensuring that administrators and pharmacists are presented only with the functions available to them.

The final stage of development consisted of testing and validation conducted in three phases, each addressing a distinct evaluation concern that corresponds directly to the problem identified in this study.

The first phase employs **black-box testing**, a functional testing approach that evaluates system behavior based on inputs and expected outputs without examining the internal code structure. Black-box testing is effective in identifying functional and usability issues and is appropriate for verifying whether each feature of the system produces correct and expected results under defined conditions [19]. For each feature, including login, inventory management, CSV import, expiration monitoring, and report generation, a structured test case table is prepared containing the test scenario, input, expected output, actual output, and a pass or fail result, providing a documentable and verifiable record of system correctness before proceeding to user-based evaluation.

The second phase addresses the core problem of this study, which is the inefficiency of manual medicine tracking among pharmacy staff. This phase employs **task-based usability testing** grounded in the ISO 9241-11:2018 standard, which defines usability as the extent to which a system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use [20]. In this phase, evaluators are asked to perform specific inventory-related tasks using PharmaTrack, such as identifying medicines nearing expiration and locating low-stock items, which are the exact tasks that pharmacy staff currently perform manually. Effectiveness is measured by whether the evaluator successfully completes each assigned task, and efficiency is measured by the time taken to complete it. These results are then compared against a manual condition using the same dataset presented in a spreadsheet format, so that the improvement in task performance attributable to PharmaTrack can be directly quantified and used to address whether the system reduces the inefficiency identified in the problem statement.

The third phase captures the subjective experience of users through a **satisfaction questionnaire** administered after the task-based evaluation. The questionnaire is grounded in the ISO/IEC 25010:2011 software quality model, which provides a comprehensive framework for evaluating software products and has been applied in existing research as an objective, standardized approach to minimizing subjective factors in software quality assessment [21], [22]. For this study, the evaluation focuses on three quality characteristics most relevant to PharmaTrack's scope: functional suitability, which assesses whether the system correctly performs its intended inventory and expiration monitoring functions; usability, which evaluates how effectively and easily the two user roles can interact with the system; and reliability, which measures the system's ability to perform consistently under the test conditions. Responses are collected using a five-point Likert scale, a widely adopted measurement instrument for quantifying user attitudes and perceptions first introduced by Likert in 1932 [23]. Each criterion is rated from 1 (Strongly Disagree) to 5 (Strongly Agree), and mean scores are computed and interpreted using a descriptive rating scale to determine the overall level of user satisfaction and system acceptability.

Together, the three phases form a complete and cohesive evaluation framework. Black-box testing confirms that the system functions correctly, task-based usability testing measures whether the system concretely improves the efficiency of pharmacy staff in tracking medicine stocks and expiration dates, and the satisfaction questionnaire captures the overall user perception of the system's quality.

1. **Requirement Analysis** – Identify inventory, role-based access, and reporting requirements.
2. **Source Code Review** – Analyze the original open-source pharmacy management system by Varshini E.
3. **System Customization** – Remove unrelated modules and implement role-based access control.
4. **Bulk Import Implementation** – Develop a CSV file upload and parsing feature for bulk medicine data migration.
5. **Database Enhancement** – Populate the MySQL database with imported and manually entered medicine datasets.
6. **Interface Adjustment** – Modify the interface to reflect administrator and pharmacist access levels.
7. **Testing and Validation** – Test inventory accuracy, role restrictions, and bulk import functionality using simulated data.

3.7 Source Code Attribution and Modification

This study is based on an open-source web-based pharmacy management system developed by **Varshini Elangovan**, implemented using PHP and MySQL. The original system was designed to manage pharmacy inventory, customer records, employee information, suppliers, purchases, and sales.

For this research, the source code was modified to align with the objectives of inventory and expiration monitoring. Several components unrelated to the study scope were removed, while new functionalities, such as bulk data migration and expanded inventory datasets were added. These modifications ensure that the resulting system reflects the specific goals of this study while maintaining proper attribution to the original developer.

In addition to feature removal and enhancement, a bulk data import mechanism was implemented to support efficient migration of medicine records. This feature was not part of the original system and was developed specifically for this study. User roles were also redefined to align with real pharmacy workflows, ensuring appropriate access control between administrators and pharmacists.

This chapter presented the methodology used in developing PharmaTrack as a standalone, web-based pharmacy inventory and expiration monitoring system. By adapting an existing PHP-MySQL open-source application, the study demonstrates how targeted system modification and dataset enhancement can produce a functional prototype suitable for academic evaluation. The methodology emphasizes simplicity, correctness, and alignment with real-world pharmacy inventory needs while avoiding integration with live POS systems.

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